

Improper integrals – 1

Convergence: the two scales and comparison

Homework

12 сентября 2026 года

Theoretical minimum. An improper integral is a limit, and an integral with several singularities is split so that each piece carries exactly one. Two scales govern everything: x^{-p} at infinity converges for $p > 1$, $(x - c)^{-p}$ at a singular point converges for $p < 1$, with logarithmic refinements in both cases. Comparison works for one-signed integrands only. The Cauchy criterion is the general tool, and absolute convergence implies convergence. For monotone $f \geq 0$ the integral, the series $\sum f(n)$ and the condensed series $\sum 2^k f(2^k)$ live or die together. Convergence of the integral says nothing about $f(x) \rightarrow 0$ unless f is uniformly continuous or monotone; the correct general statement is $\liminf x f(x) = 0$.

1. Determine for which real p and q the integrals $\int_0^1 x^p |\ln x|^q dx$, $\int_2^\infty \frac{dx}{x^p \ln^q x}$ and $\int_0^\infty \frac{x^p}{1+x^q} dx$ converge.
2. Prove that $\int_0^\infty \frac{dx}{1+x^p \sin^2 x}$ converges exactly for $p > 2$, and explain why the naive comparison with x^{-p} gives the wrong answer.
3. (VJIMC 2017) Let $f: (1, \infty) \rightarrow \mathbb{R}$ be continuously differentiable with $f(x) \leq x^2 \log x$ and $f'(x) > 0$ for every $x \in (1, \infty)$. Prove that $\int_1^\infty \frac{dx}{f'(x)} = \infty$.
4. (MexMat, 2011) Let $h: [1, \infty) \rightarrow [1, \infty)$ be a continuous strictly increasing surjection. Prove that

$$\int_1^\infty \frac{h(t)}{t^2} dt = 1 + \int_1^\infty \frac{ds}{h^{-1}(s)},$$

provided at least one of the two integrals is finite.

5. Prove that if $f \geq 0$ is non-increasing and $\int_0^\infty f$ converges, then $xf(x) \rightarrow 0$; construct a continuous $f \geq 0$ with convergent integral which is unbounded, and another one for which $\limsup f(x) = 1$.
6. Prove that for continuous $f > 0$ on $[1, \infty)$ with $\int_1^\infty f < \infty$ one always has $\liminf_{x \rightarrow \infty} xf(x) \ln x = 0$, and show that $\ln x$ cannot be replaced by $\ln x \ln \ln x$ in general.
7. Prove the integral test in the form $|\sum_{k \leq n} f(k) - \int_1^n f| \leq f(1)$ for monotone f , and construct a continuous $f \geq 0$ for which $\int_1^\infty f$ converges while $\sum f(n)$ diverges, and another one where the opposite happens.
8. Let f be continuous on $[0, \infty)$ and suppose $\int_0^\infty f$ converges. Prove that $\int_0^\infty f(x)e^{-x}dx$ converges as well, and decide whether $\int_0^\infty f(x)^2dx$ must converge when $f \geq 0$.
9. Investigate the convergence of $\int_0^1 \frac{dx}{x^p |\sin \frac{1}{x}|^q}$ and of $\int_1^\infty \frac{\ln(1+x^a)}{x^b} dx$ for all values of the parameters.
10. Let f be continuous on $[0, \infty)$ and let $\int_0^\infty f$ converge. Prove the integral form of Kronecker's lemma: $\frac{1}{T} \int_0^T xf(x) dx \rightarrow 0$ as $T \rightarrow \infty$. Does the conclusion survive if the convergence of $\int_0^\infty f$ is replaced by boundedness of its partial integrals?